

DBT TRAINING

● Beginner · Module 06 · 90 min

Testing Data Quality

Not optional. CI rejects Silver and Gold PRs that are missing mandatory tests on key columns.

Why Tests Are Mandatory

Without tests, transformations break silently

- A column renamed in Bronze → staging model silently returns NULLs
- Duplicate surrogate keys → Power BI relationships corrupt
- NULL in a required column → propagates downstream undetected
- A Gold mart shows wrong numbers → business discovers it weeks later

The failure is always silent. The discovery is always late. The cost is always high.

Tests are a CI requirement

Silver and Gold PRs with missing tests on `_key` columns → **rejected**

Missing FK tests on fact tables → **rejected**

Zero tests on a new Silver model → **rejected**

Reference: `dbt-test-strategy` skill — full placement guide per layer

Two Types of dbt Tests

Generic tests — 95% of what you'll write

Defined in `schema.yml`. Reusable. Parameterised.

```
models:
  - name: fct_prescription
    columns:
      - name: prescription_key
        tests:
          - unique
          - not_null
      - name: patient_key
        tests:
          - relationships:
              to: ref('dim_patient')
              field: patient_key
```

Singular tests — for complex business rules

Standalone `.sql` files in `tests/`. Return rows on failure. Two common patterns:

Pattern 1 — business rule on aggregates (`GROUP BY ... HAVING`):

```
-- tests/assert_fct_revenue_no_negative_amounts.sql
SELECT customer_key, SUM(amount_net) AS total
FROM {{ ref('fct_revenue') }}
GROUP BY 1
HAVING total < 0
```

Pattern 2 — orphan FK check (`LEFT JOIN ... WHERE IS NULL`):

```
-- tests/assert_no_orphan_prescriptions.sql
SELECT p.prescription_key
FROM {{ ref('fct_prescription') }} p
```

The Four Generic Tests

unique

```
- name: prescription_key
  tests:
    - unique
```

Fails if any value appears more than once.
Required on all

`_key` columns.

not_null

```
- name: prescription_key
  tests:
    - not_null
```

Fails if any value is NULL. Required on all `_key` columns.

Test Severity and `dbt build`

Severity: `error` vs `warn`

Global default — `dbt_project.yml`:

```
tests:
  analytics:
    +severity: warn    # project-wide default
  silver:
    +severity: error   # Silver always errors
  gold:
    +severity: error   # Gold always errors
```

Per individual test — `schema.yml`:

```
- name: prescription_key
  tests:
    - unique:
        config:
          severity: error # ← CI stops, pipeline halts
- name: dosage_amount
  tests:
    - not_null:
```

`dbt build` — the only correct command

```
# ❌ WRONG
dbt run && dbt test

# ✅ CORRECT — always
dbt build
```

`dbt build` runs models and tests in **DAG order**. If `dim_patient` fails a test, `fct_prescription` (which depends on it) is **never built**.

`dbt run && dbt test` runs all models first — loading bad data into `fct_prescription` before the tests even run.

Never use `dbt run && dbt test` in CI. Always `dbt build`.

Mandatory Test Requirements

Column type	Required tests
<code>_key</code> columns (surrogate PKs)	<code>unique</code> + <code>not_null</code> (both, always)
FK columns (referencing another model)	<code>relationships</code>
Status / type columns	<code>accepted_values</code> (if finite value set exists)
Business-critical measures	<code>not_null</code>

CI check: Silver and Gold models must have at least one test per `_key` column. A PR with zero tests on a new Silver model will not pass review. This is checked manually via the `dbt-test-strategy` skill until CI automation is complete.

Full test placement guide (when to test in Bronze vs Silver vs Gold, `store_failures` config, `dbt-expectations`) → `dbt-test-strategy` skill

Exercise: Write Tests for `fct_prescription` (30 min)

The model has no tests. Add the complete test suite to `schema.yml`.

Column	Type	Notes
<code>prescription_key</code>	Surrogate PK	MD5 hash
<code>patient_key</code>	FK → <code>dim_patient</code>	—
<code>doctor_key</code>	FK → <code>dim_doctor</code>	—
<code>prescription_date</code>	Date	Required
<code>medication_type</code>	String	<code>tablet</code> , <code>liquid</code> , <code>injection</code> , <code>topical</code>
<code>dosage_amount</code>	Number	Can be null — not yet confirmed

MODULE 06 COMPLETE

Next: Module 07

Documentation — the final Beginner module

Prep Q1: Generic test vs singular test – difference?

Prep Q2: Name the four built-in generic tests

Prep Q3: What does dbt build do differently from dbt run && dbt test?

Prep Q4: Two mandatory tests on every _key column in Silver?